

AUTOMATED EXPENSES LOGGER

USING n8n

Overview

This workflow automates expense tracking by allowing users simply to send an image or PDF of an invoice or receipt through Telegram, and the bot processes it automatically by extracting the key information such as name, amounts, and dates.

Problem

Manual entering and tracking of expenses are time consuming and, especially when handling multiple receipts, and invoices.

Solution

Created an automated workflow that captures users' invoice or receipts from image or PDF format through Telegram, and the bot automatically processes it and it will log the key details from the users' input into a Google Sheet.

Workflow Breakdown

1. Telegram bot receives users' input (Image or PDF).
2. Telegram bot sends a message that it will process the input.
3. The workflow will filter if the user input is an Image or PDF.
4. Extract key details from the input.
5. Logs the key details in the prepared Google Sheet
6. Send a notification to the user that their input is has been processed.

Tools & Integration

1. n8n
2. Telegram
3. Google Sheets
4. Google Gemini 2.5 Flash
5. Mistral AI OCR API

Key Features

1. Real-time automation
2. No manual work
3. Custom logic (switch/loop)

Results & Impact

1. Reduced manual logging and tracking of expenses from minutes to seconds
2. Saved hours of work
3. Reduced human errors

Challenges

At the beginning of building the workflow, I encountered several issues, especially around selecting the right OCR API. I tested different OCR tools, but they did not produce reliable results.

I also tried using an AI agent with a prompt to extract data from images and PDFs. It worked initially, but it became unstable and eventually stopped producing consistent results.

After hours fixing and researching, I discovered MISTRAL AI OCR API, read its documentation, and integrated it into the workflow. This solved the OCR extraction issue.

Another challenge that I faced was looping behavior in the workflow. The bot was sending multiple Telegram messages based on the number of items in the invoice. For example, if an invoice had 6 items, the bot would send 6 separate notifications.

To fix this, I used a summarize node to count the items in the loop and properly control the output before sending it to Telegram. This resolved the issue and the workflow now runs correctly.

Learnings

Learned how to use right tools and nodes for certain tasks.

Future Improvements

1. Accepts invoices and receipts in different formats other than pdf and images.
2. Voice message in the Telegram bot.
3. A dashboard automatically created.

CONCLUSIONS

This project shows how automation with n8n can turn manual processes into fast, reliable solution that drives result.